School or Department: SEG

Qualification Type: PET (AY2024 Semester 2)

Module Code: E211

Cohort: Main

Assessment: MUE

Assessment Document:

Assessment Paper AY 2024 Semester 2 E211 MUE Main (1)

Section A (12 Questions)

Q1. Which of the following is NOT a correct characteristic of a **M/D/3** queuing model?

Number of channels is 3

Number of service stages is 3

Service time is constant

Arriving population behaviour is patient

Q2. To find a delivery route for the network in Table 1, the Nearest Neighbour Method is applied. The routing solution with **starting node at C** would be:

Table 1: Travel Distance between Nodes

	A	B	C	D
A		8	10	12
B	8		15	20
C	10	15		13
D	12	20	13	

A – B – C – D – A

C – D – A – B – C

C – A – B – D – C

C – A – B – A – D

Q3. Alicia is tasked to forecast the demand for a new product. She brings together a group of participants for them to discuss and share their views on the new product. This form of forecasting is known as:

Market research

Grassroots forecasting

Delphi method

Focus group

Q4. Which of the following is an advantage of using simulation modelling to study a large-scale production process?

It provides precise solutions to all production scenarios.

It allows the study of complex processes without disrupting the actual production.

It eliminates the need for expertise or understanding of the production system.

It reduces the implementation costs of any proposed solution.

Q5. In FlexSim 3D model, to model the packing of carton boxes onto a pallet, which object should you be using?

Processor

Separator

Combiner

Multiprocessor

Q6. Which of the following is NOT an output from Experimenter results?

State Pie Chart

Scenario Comparison

Replications Plot

Data Summary

Q7. Figure 1 shows the options for Arrival Spacing at the Arrivals table of FlexSim Healthcare. Which of the following statements is FALSE?

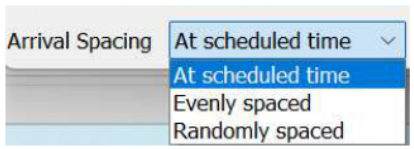


Figure 1: Arrival Spacing Options

If 'At scheduled time' is selected, the people will arrive at the start of the specified period.

To model random arrival of people within a period, 'Randomly spaced' should be selected.

'Evenly spaced' should be selected to spread out the arrivals of people evenly within the specified period.

It is possible to select a different option for each row of the Arrivals table.

Q8. Palantial is a local company that provides supplies from its warehouse to 6 other locations in the country. The distribution network for its products is shown in Figure 2a. The nodes represent locations and values on the arcs represent distances between locations.

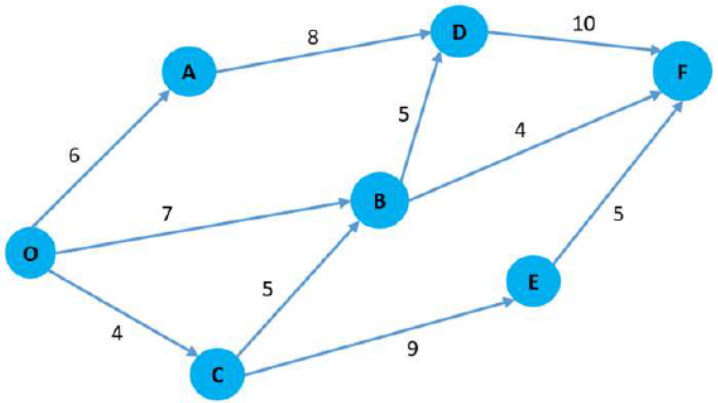


Figure 2a: Palantial's Distribution Network

Q8a. Table 2a shows the partial workings to find the shortest path from the warehouse (node O) to each of the other locations (Nodes A to F) in the network.

Table 2a: Partial Workings for Shortest Path Problem

Iteration	Solution Set	Arcs	Total Distance
Iteration 1	O	O → A O → B O → C	6 7 4
Iteration 2	O C (4)	O → A O → B C → B C → E	6 7 **V** 9 13
Iteration 3	O C (4) A (6)	O → B C → B C → E A → D	7 **V** 9 13 14
Iteration 4	O C (4) A (6) B (7)	C → E A → D B → D B → F	13 14 12 11
...	...	...	...

Q8a(i). Name the algorithm used in Table 2a to solve the shortest path problem.

Dijkstra's algorithm

Q8a(ii). Calculate the value of **V** in Table 2a.

V = 4+5 = 9

Q8a(iii). Which node should be added to the solution set at **Iteration 5**? Explain.

Node F as it has the next shortest distance of 11.

Q8a(iv). Work out the shortest path from node O to node D <u>and</u> state the total distance of the shortest path from O to D.

Shortest path: O -> B -> D. Total distance = 7+5 =12

Q8b. A manufacturer has a plant (Node A) producing a medical device. The product is shipped to customers (Nodes D & E). The distribution network for shipping the product is shown in Figure 2b where the values on the arcs represent unit shipping costs. The manufacturer wants to find out the lowest cost shipping plan by formulating a Linear Programming (LP) model to solve the Minimum Cost Flow problem.

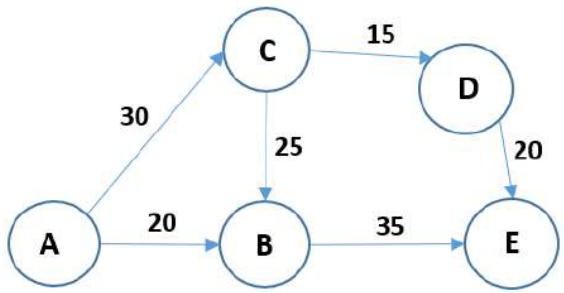


Figure 2b: Medical Device Distribution Network

The decision variable Xij is defined as the quantity of products to ship from node i to node j. (i, j = A, B, C, D or E).Given:

- Node A can supply up to 200 units
- Nodes D and E have demands of 125 and 75 units respectively
- Nodes B and C are transhipment hubs with zero supply and demand.

Q8b(i). Formulate the objective function of the LP model.

Minimize total shipping cost, Z = 20XAB + 30XAC +25XCB + 15XCD + 35XBE + 20XDE

Q8b(ii). Formulate the following constraints:

1. Net flow constraint for **Node B**.
2. The maximum shipping capacity from Node D to E is 30.
3. Net flow constraint for **Node D if its demand is increased to 150**.

1. Node B: XBE - XAB - XCB = 0
2. XDE <= 30
3. XDE - XCD >= -150

Q8b(iii). Is it necessary to add the non-negative decision variable constraint to the LP model? Explain.

Yes, as the result will change if we do not include non-negativity constraints when doing LP.
Yes, it is necessary. The decision variables represent physical quantities of products being shipped, which cannot be negative in the real world (Xij >= 0).

Q8c. A product is to be delivered from Depot (O) to five other stores (A, B, C, D and E). A van will leave the Depot (O), visit each store once and return to the Depot after delivery. Travel distances between locations are given in Table 2b.

Table 2b: Travel Distances Between Locations

	O	A	B	C	D	E
O	-	31	19	35	21	14
A	31	-	13	33	23	6
B	19	13	-	24	26	25
C	35	21	24	20	15	18
D	33	23	26	15	-	25
E	34	6	25	18	25	-
F	36	11	21	32	15	18

Clark and Wright savings method is used to plan the delivery route. Table 2c shows the **Savings** in distance (km) between 2 stores ranked in descending order.

Table 2c: Savings Ranked in Descending Order

Rank	Savings	From	To
1	41	C	D
2	39	A	E
3	37	A	B
4	33	A	C
5	31	C	E
6	30	B	C
7	29	A	D
8	14	B	D
9	10	D	E
10	X	B	E

Q8c(i). Calculate the savings between nodes B and E (labelled 'X' in Table 2c).

Savings = 19+14-25 = 8

Q8c(ii). Referring to Table 2c, at Rank 1, a sub-tour of O-C-D-O is constructed. Explain why at the next step (Rank 2) a new sub-tour O-A-E-O should be constructed.

Neither A or E has been assigned to a route, so we initiate a new route including both A and E.

Q8c(iii). Work out the complete delivery route based on Clark and Wrights Savings Method. Show your workings starting at Rank 3.

Rank 3: O-C-D-O, O-B-A-E-O
Rank 4: skip (A is not adjacent to O)
Rank 5: O-B-A-E-C-D-O (Join C & E; Merge the 2 tours)

Q8c(iv). Does Clark and Wrights Savings Method guarantee an optimal solution when applied to routing problems?

It does not guarantee an optimal solution.

Q9. *Show your workings clearly and round off your answer to* ***2 decimal places*** ***where necessary.***

Danial works in a company that develops consumer smart watches. His role in the company is sales and demand planning.

Q9a. The company launched a smart watch some months ago. Danial would like to predict future customer demand based on past sales data (given in Table 3a).

Table 3a: Historical Sales Data

Month	Sales Figure (Units)
1	180
2	185
3	222
4	217
5	269
6	279

Q9a(i). Calculate the forecast for Month 7 based using **3-month weighted** moving average method with weights of 3 (most recent), 2, 1 (least recent).

Forecast (month 7) = ((279*3)+(269*2)+(217*1))/(3+2+1) = 265.33

Q9a(ii). Based on the workings in Table 3b, calculate the forecast for Month 7 using Holt's method with smoothing constants α = 0.7 and β = 0.4.

Table 3b: Holt's Method Workings

Month	Sales Figure	L_t	T_t
		151.13	21.20
1	180		
2	185		
3	222		
4	217		
5	269	260.32	24.09
6	279	280.62	Y

T6 = 0.4(280.62-260.32)+(1-0.4)(24.09) = 22.57
H7 = L6 + mT6
= 280.62 + (1)(22.57)
= 280.62 + 22.57 = 303.19

Q9a(iii). Which of the above forecasts is more accurate if the sales in Month 7 turns out to be 295? Justify your answer by comparing the Mean Absolute Deviation (MAD) of the 2 forecasts.

WMA AD: |295 - 265.33| = 29.67
Holt's method AD: |295 - 303.19| = 8.19

Holt's method is more accurate due to having smaller forecasting errors (8.19) as compared to WMA AD (29.67).

Q9b. Danial wants to know how the advertising budget affects the sales of a smart watch. Figure 3 plots the smart watch sales against the corresponding advertising budget. A linear regression line is plotted.

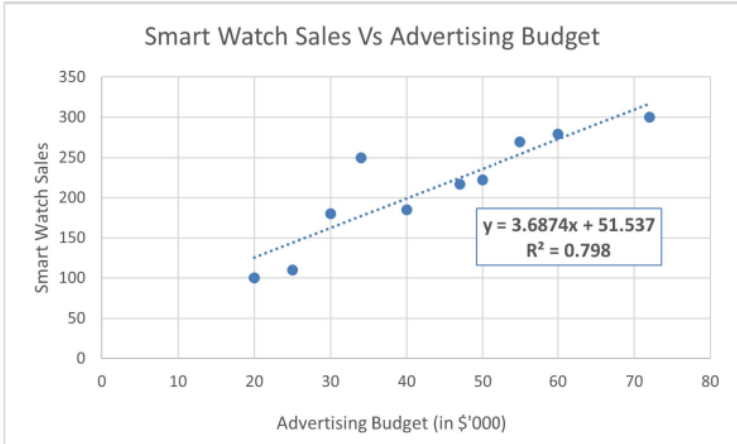


Figure 3: Scatter plot of Smart Watch Sales Vs Advertising Budget

Q9b(i). Identify the independent variable and dependent variable in the linear regression model.

Independent variable - Advertising Budget

Dependent variable - Smart Watch Sales

Q9b(ii). Calculate the correlation coefficient R and comment on the correlation between smart watch sales and the advertising budget.

Correlation coefficient R = SQRT(0.798) = 0.89 (positive correlation)
There is a strong positive correlation between smart watch sales and advertising budget.

Q9b(iii). Predict the sales of the smart watch if the advertising budget is $65 ('000).

y = 3.6874(65) + 51.537 = 291.22

Q9c. Danial has developed an Excel template for the Multiplicative Decomposition Method (Table 3c) to conduct a study a smart watch that exhibits seasonal demand pattern.

Table 3c: Workings of Multiplicative Decomposition Method

Year/ Quarter	Sales Figure (Y)	Time Period (X)	Moving Average	Centred Moving Average	Seasonal Ratio
2024/ 1	630	9	385		
2024/ 2	310	10	391.25	'A'	'B'
2024/ 3	285	11	430		
2024/ 4	340	12	449.25		

Q9c(i). Calculate the Centred Moving Average for **2024 Quarter 2** (labeled 'A' in Table 3c).

A = (385+391.25)/2 = 388.13

Q9c(ii). Calculate the Seasonal Ratio for **2024 Quarter 2** (labeled 'B' in the Table 3c).

B = 310/388.13 = 0.7987 = 0.80

Q9c(iii). Given that the Seasonal Index for Quarter 2 is 0.86, compute the Deseasonalized Value for **2024 Quarter 2**.

Deseasonalized value = Sales figure/Seasonal index
= 310/0.86 = 360.47

Q9c(iv). Given that the linear trend for deseasonalized values can be represented by the regression model: Y = 170.98 + 17.37X, compute the seasonalized sales forecast for **2025 Quarter 2**.

Deseasonalized forecast = 170.98 + 17.37(14) = 414.16
Seasonal index = 0.86
Seasonalized sales forecast = 414.16*0.86 = 356.18

Q10. *Show your workings clearly*. ***Round off your answers to 2 decimal places where necessary.***

In a supermarket, an average of 75 customers enter the self-checkout area every hour according to a Poisson process. There are 5 self-checkout machines available. Each customer spends an average of 3 minutes completing the checkout. The checkout time follows a negative exponential distribution.

Q10a. Calculate the customer arrival rate (in per minute) at the self-checkout area.

Arrival rate = 75/60 = 1.25 per min

Q10b. Calculate the utilization of the machines at the self-checkout area.

Service rate = 1/3 = 0.333 per min
Utilization = 1.25/(5*0.333) = 0.75

Q10c. Calculate the minimum number of machines required at the self-checkout area for the queuing system to be in steady state condition.

Arrival rate = 1.25 per min, Service rate = 0.333 per min

Steady state equilibrium holds when arrival rate < k(service rate) [for k servers]
Therefore, minimum k > 1.25/0.333 > 3.75
Minimum number of machines required, k = 4

Q10d. Given that average number of customers in queue is 1.3854, calculate the average waiting time in queue.

Wq = Arrival rate/Lq
= 1.25/1.3854
= 0.90

Wq = Lq/Arrival rate
= 1.3854/1.25
= 1.10832

Q10e. Using Table 4 where n is the number of customers in the system, calculate the probability that there is **at least 1 customer in the queue**. The number of self-checkout machines is 5 (value of k).

Table 4: Partial Workings of M/M/5 Queuing Model

n	$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	0.019	0.486
1	0.070	0.365
2	0.131	0.274
3	0.164	0.205
4	0.154	0.154
5	0.115	0.115
6	0.072	0.087
7	0.039	0.065
8	0.018	0.049
9	0.008	0.037

0.019 + 0.070 + 0.131 + 0.164 + 0.154 + 0.115 = 0.653
P(n>=6) = 1 - P(n<=5)
= 1 - 0.019 - 0.070 - 0.131 - 0.164 - 0.154 - 0.115
= 0.347

Q10f. The supermarket also runs a cashier checkout.

Q10f(i). In order for a M/G/1 model to be applied to study the queuing system at the cashier checkout, what assumptions are required for the following?

1. Customer behaviour
2. Queue capacity

1. Patient

2. unlimited

Q10f(ii). On a particular day, the arrival rate at the cashier checkout is 4.8 per minute. The average time to check out is 2.5 minutes with a standard deviation of 1 minute. Calculate the average waiting time in queue.

Arrival rate = 4.8 per min
Service rate = 1/2.5 = 0.4 per min
Avg waiting time in queue (Wq) = Lq/Arrival rate = -1.582 mins

Q10f(iii). How will the performance of the M/G/1 queuing system change when the standard deviation of the service times decreases? Explain.

Lq & Wq will decrease while Ls and Ws will increase.
As the standard deviation service time (standard deviation = 0) of service times gets smaller, the performance of the queuing system becomes better → waiting time and queue length become shorter.

Q11. Figure 5a shows a simulation snapshot of a manual car wash facility. The flow items (boxes) represent cars. Incoming cars are serviced at either Bay1 or Bay2.

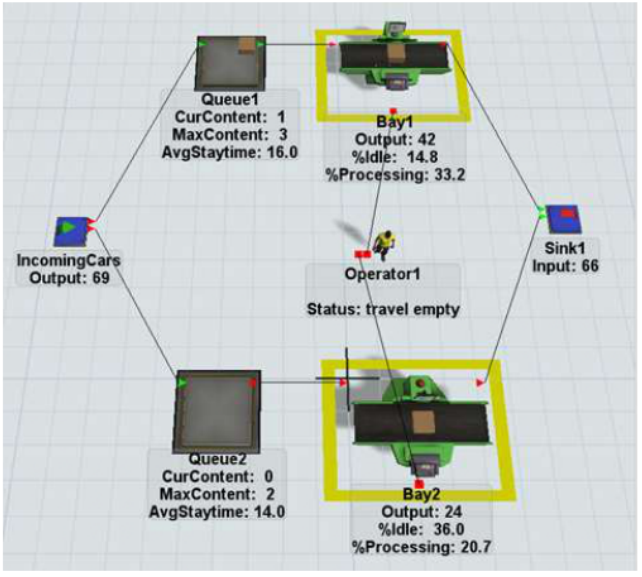


Figure 5a: FlexSim Car Wash Simulation

Q11a. Referring to Figure 5a, state the FlexSim 3D object used to model:

i) IncomingCars - source

ii) Bay2 - processor

Q11b. Referring to Figure 5a, state the following statistics:

i) Average waiting time for Bay1 - 16 (AvgStayTime)

ii) Number of cars that have been serviced at the facility - 66 (Sink)

Q11c. Referring to Figure 5a, the statistics under Bay1 show that it is 14.8% idle and 33.2% processing. What state might Bay1 be in for the remaining 52% of the time?

Processing if there is many cars.

Remaining 52% of the time, Bay1 is most likely in the Waiting for Operator state.

Q11d. In 'IncomingCars' object properties, an On Creation trigger was created as shown in Figure 5b. Describe what this trigger does.

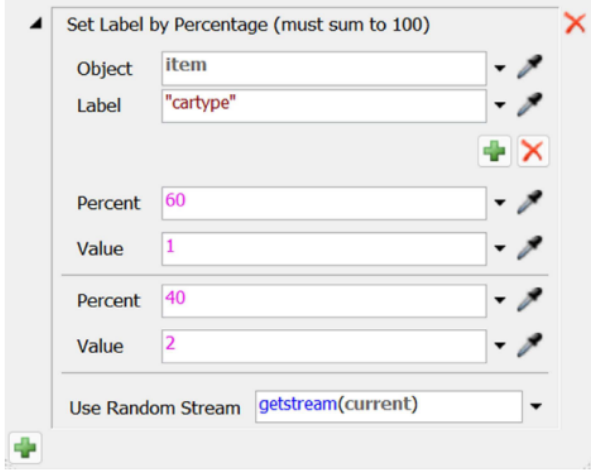


Figure 5b: On Creation Trigger Settings

60% will go to queue1 (bay1)
40% will go to queue2 (bay2)

The trigger creates a label called 'cartype' on each incoming flow item (car).
It randomly assigns a value of 1 to 60% of the cars, and a value of 2 to the remaining 40%.

Q11e. Operator1 is a task executer required to perform the manual car wash. What are the 2 things to model for Operator1 so that his actions are realistic?

Connect operator1 to bay1 and bay2 via center port connection. ✓
Reset the simulation and re-run again to see. ✗
Inside the properties window for both Bay1 and Bay2, check the box for "Use Operator(s)" and specify the number of operators needed (1). Without this, the bays won't actually call him to do the work

---

Q12. *Round off your answer to **2 decimal places** where necessary.*

Tina has built a FlexSim Process Flow model to study a production line comprising a Machining process and a QC process. Machining is carried out at Work Station whereas QC is done at QC Station.

Q12a. Figure 6a shows a snapshot of the simulation model. Answer the following questions:

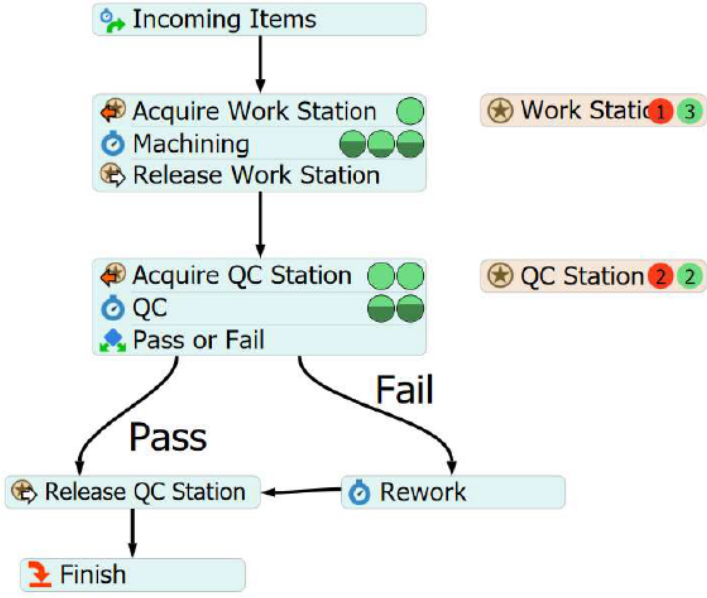


Figure 6a: FlexSim Process Flow Model

i) How many items are there currently in the production line?

3 ✗ 8

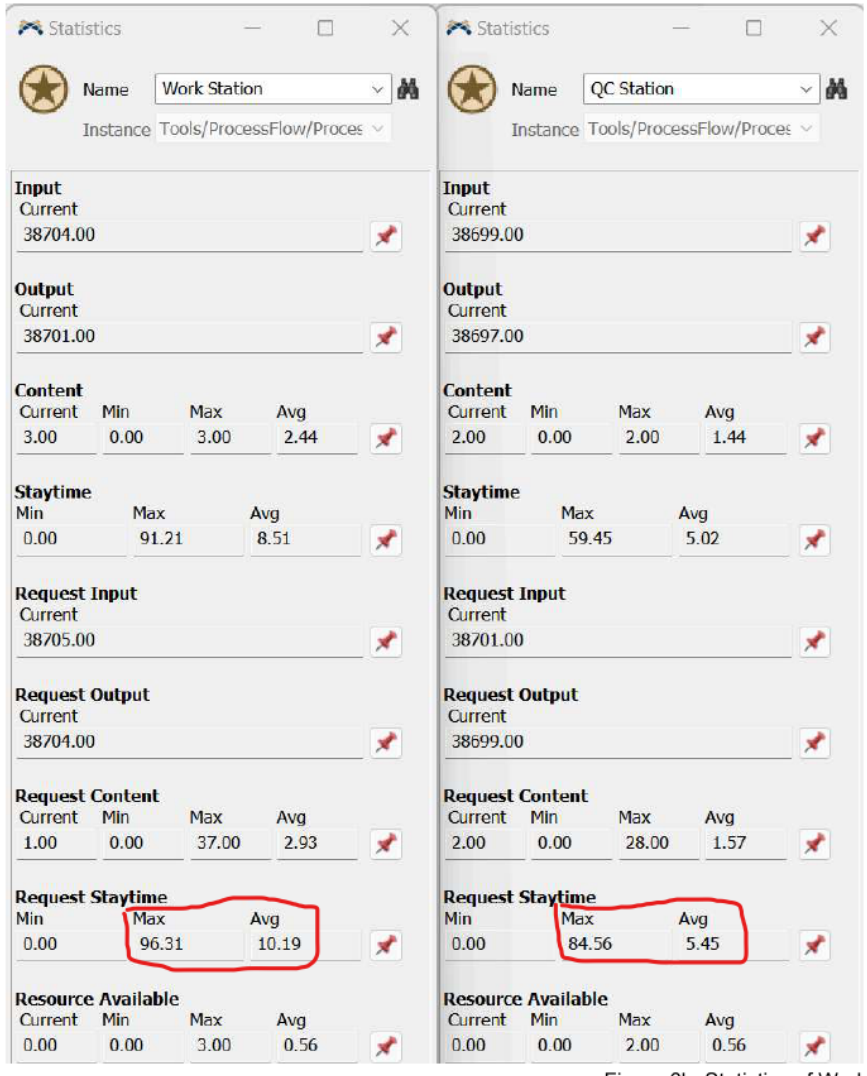
ii) How many Work Stations are there?

2 (Work + QC stations) ✗ 3

iii) State the FlexSim process flow activity used to model 'Rework'.

Delay ✓

Q12b. Figure 6b shows the statistics of Work Station and QC Station at the end of the simulation.



Statistics

Name: Work Station

Instance: Tools/ProcessFlow/Proces

Input Current	38704.00			
Output Current	38701.00			
Content	Current 3.00	Min 0.00	Max 3.00	Avg 2.44
Staytime	Min 0.00	Max 91.21	Avg 8.51	
Request Input Current	38705.00			
Request Output Current	38704.00			
Request Content	Current 1.00	Min 0.00	Max 37.00	Avg 2.93
Request Staytime	Min 0.00	Max 96.31	Avg 10.19	
Resource Available	Current 0.00	Min 0.00	Max 3.00	Avg 0.56

Statistics

Name: QC Station

Instance: Tools/ProcessFlow/Proces

Input Current	38699.00			
Output Current	38697.00			
Content	Current 2.00	Min 0.00	Max 2.00	Avg 1.44
Staytime	Min 0.00	Max 59.45	Avg 5.02	
Request Input Current	38701.00			
Request Output Current	38699.00			
Request Content	Current 2.00	Min 0.00	Max 28.00	Avg 1.57
Request Staytime	Min 0.00	Max 84.56	Avg 5.45	
Resource Available	Current 0.00	Min 0.00	Max 2.00	Avg 0.56

Figure 6b: Statistics of Work Station and QC Station

Q12b(i). Determine the bottleneck of the process by comparing the waiting times for the 2 Stations.

[We look at request staytime] Work Station: 96.31, QC Station: 84.56

Hence the bottleneck will be Work Station.

Q12b(ii). Calculate the utilization of the Work Stations.

Avg utilization = Average content/Max Resource = 2.44/3 = 0.81

Q12c. Tina wishes to build a 3D simulation model of the production line. List 2 additional information/ data that she would need to build the 3D simulation model in FlexSim.

Add enter zone (after incoming items) and exit zone (after releasing QC station).
Create performance measures

Physical Layout & Distances: The actual dimensions of the facility floor plan and the distance between the Work Station and QC Station.
Transportation Speeds: The walking speed of operators or the speed of the conveyors moving items between the stations.

Close

Powered by AvePoint, Inc